

Research - AI tools and technologies

AI tools

- Conversational chatbots: Having chatbots for the digital solution would allow users to make a conversation with the chatbot to be able to communicate and be provided the needed information. Using chatbots onto the website, users can receive instant answers and can be asked anytime without waiting.
- Screen reader - a software program that converts text and images into synthesized speech or a Braille display. The screen reader would use text-to-speech synthesizers to read the text aloud for the users by using human-like speech from the digital text. This is mainly used as an accessibility feature to people who prefers audible speeches over reading text and people with sight loss.
- Screen magnifiers - tools that help enlarge text, images and cursors on computers and phones. This is another feature for users to be able
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- Braille displays - hardware devices that convert digital text into tactile braille, allowing users to read via touch.

Technologies

- Machine learning - uses data and algorithms for computers to identify patterns and make predictions.
- Natural Language Processing - a branch of artificial intelligence that enables computer to understand and generate human language.
- Generative AI - a type of artificial intelligence that generates new content
- Robotics and Automation - using technology to make machines perform tasks without human intervention
- Deep Learning - a subset of machine learning

How AI-generated data differs from human-generated data

AI - generated data

- Highly structured and consistent data
- Known as synthetic data
- Fast and cheap
- Examples: audio and speech generation, financial and transaction records, text and language data
- Produced by automated systems and algorithms
- Raises ethical and security concerns
- Beneficial for AI applications

Human-generated data

- Suitable for personal anecdotes and sentiments
- Examples: text files, media, social media, feedback
- Expensive and time consuming to collect and process data
- Limited by human capacity and availability
- Content created by people through daily writing and social interactions.
- Examples: questionnaires, speech, emails,
- Contains errors and biases
- Reflects human behaviour and emotions

Societal and organisational implications of AI integration

Societal implications of AI integration

AI integration brings many benefits in productivity and decision-making in companies as well as raising challenges in ethical and privacy concerns.

Economic and employment impacts

AI enhances productivity and efficiency across industries however it also disrupts traditional employment, leading to potential displacements for employees and requiring new skill sets. While AI might be able to open new opportunities in tech sectors, the transition can lead to economic inequality as employers are unable to meet the requirements of the high -skill roles.

Trust in AI

Having biased algorithmic decisions can erode trusts in AI, leading to a negative reputation and decline in the use of the technologies. As AI technologies may offer future opportunities, people are wary of risks of using AI.

People who are unfamiliar with AI technologies may be skeptic about AI performing tasks instead of human beings, fearing that they might be replaced of certain job roles such as software engineer. With the lack of trust in AI, there will be resistance to AI integration and reduced engagement of using AI.

Along with skepticism about AI, algorithmic bias can occur when AI systems produce discriminatory outcomes from flaws in AI systems, leading to less accuracy and limited performance from flawed algorithms.

Organisational implications of AI integration

Workplace culture and human connection: Excessive reliance on AI can weaken interpersonal relationships as employees increasingly interact with AO rather than colleagues leading to a loss of purpose at work. Employees should have balance of human interactions and AI intelligence so companies should pursue hybrid intelligence to maximise AI's benefits.

Cognitive skills and expertise: Overuse of AI could threaten cognitive skills and learning. Delegating too many tasks to AI would lead to cognitive decline and diminished expertise due to the passing of tasks to another tool that doesn't require any AI integration which can lead to less engagement and learning opportunities.

Accessibility

Social accessibility - designing and creating digital content for people with disabilities

Organisational accessibility - designing an organisation's physical spaces, digital systems, and workplace to be fully usable and inclusive for everyone

Feasibility assessment

This assessment is an evaluation that determines whether the project is practical or viable before major resources are committed.

Technical requirements:

- Skill analysis
- Hardware and infrastructure - evaluates the physical equipment needed for the project
- System integration
- Technical training
- Scalability and maintenance
- Performance
- Security - verify user's identity and rules of making an user account
- Interoperability - software must run compatibly on web browsers and mobile operating systems

Ethical considerations

A set of moral principles and values that guide decision-making and professional practices to ensure that the participants are protected and respectful to the public

- Human participants
- Bias and fairness
- Privacy and consent
- Anonymity
- Confidentiality
- Informed consent
- Voluntary participation
- Honest reporting - integrity

Data requirements

- Data availability

- Privacy and data protection
- Data storage
- Logistics

Potential impact

Enhanced decision-making - Analysing large data sets allows organizations to spot trends and make informed choices.

Increased productivity - AI tools automate repetitive tasks, saving more time and completing the tasks much faster

Customer service improvements - AI chatbots can enhance customer service, making the system more personalised and efficient for each user, leading to better customer satisfaction.

Resource consumption - Expanding data centres increase energy and water consumption

Ethical concerns - Biased algorithms can cause unfair outcomes in hiring.

Job displacements - automation market demands, heavily impacting on career roles.